

Assignment Title: *Case Study Analysis*

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Case Study Analysis Questions

Based on your allotted problem statement, prepare a report by answering the following questions:

- 1. Understand the Problem Statement**
 - Explain the problem in your own words.
 - Identify the objectives of the proposed system.
 - State the scope of the problem.
- 2. Identify Key Stakeholders and Challenges**
 - Identify all the stakeholders involved in the system.
 - Explain the roles and responsibilities of each stakeholder.
 - Discuss the major challenges and issues faced by the stakeholders.
- 3. Analyze the Current Approach**
 - Describe the existing system or current process.
 - Identify its strengths and limitations.
 - Analyze the problems or gaps that need to be addressed.
- 4. Propose a Suitable Solution Using Software Engineering Principles**
 - Design a software-based solution for the given problem.
 - Describe the major functional modules of the proposed system.
 - Identify the functional and non-functional requirements.
 - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
- 5. Justify the Chosen Methodology/Framework**
 - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
 - Justify why the selected methodology/framework is suitable for the proposed system.
 - Explain how it supports successful project development and maintenance.
- 6. Discuss Expected Outcomes and Limitations**
 - Describe the expected benefits of the proposed solution.
 - Explain how the solution addresses the identified challenges.
 - Discuss potential risks, implementation challenges, and limitations.
 - Suggest possible future enhancements.

Expected Deliverables

Each student should submit:

- Case Study Analysis Report
- Problem Analysis
- Stakeholder Analysis
- Current System Analysis
- Proposed Software Solution
- SDLC/Framework Justification
- Expected Outcomes and Limitations

This assignment evaluates the student's ability to **analyze a real-world problem, apply software engineering principles, select an appropriate development methodology, and design a feasible software solution**. It is suitable as an implementation-oriented case study assessment aligned with Software Engineering course outcomes.

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.

Methodology/Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.
Expected Outcomes & Limitations	Clearly explains expected outcomes, benefits, limitations, and future improvements.	Explains outcomes and limitations with minor omissions.	Limited discussion of outcomes or limitations.